Univariate Exploratory Data Analysis of MLB Dataset

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1 Data Validation

The dataset used in this analysis contains information on MLB games over the past seasons (from 1980 to 2024).

Property	Value
Number of Rows	101,135
Number of Columns	8
Missing Values	0 (0.00%)

Table 1: Dataset dimensions and quality checks for the MLB dataset.

Overall, the dataset is complete with no missing values, and column names are consistent and standardized.

2 Identify Variable Types

Each column in the MLB dataset was classified according to its variable type.

Column Name	Type	Description
season	Integer	First year of season year of the game
date	Date	Game date (converted from string)
team1	Categorical	Abbreviation for the first (home) team
team2	Categorical	Abbreviation for the second team
result	Categorical	Game outcome for team1 (W/L/T)
score1	Numeric	Points scored by team1
score2	Numeric	Points scored by team2
home_flag	Categorical	Indicates if team1 played Home or Away

Table 2: Classification of variables in the MLB dataset by type and description.

3 Descriptive Statistics

This section summarizes the statistical properties of both numeric and categorical variables in the MLB dataset. Numeric variables (`score1`, `score2`) are analyzed using measures of central tendency, dispersion, and distributional shape. Categorical variables (`result`, `home_flag`, `team1`, `team2`) are analyzed using frequency counts and proportions.

3.1 Numeric Variables

Metric	**Score1**	**Score2**
Number of Observations	101135	101135
Mean	4.592	4.467
Median	4.000	4.000
Mode	3.000	3.000
Standard Deviation	3.090	3.167
Variance	9.551	10.029
Skewness	0.957	0.989
Kurtosis	1.322	1.312
Minimum	0.000	0.000
Maximum	29.000	30.000
1st Quartile (Q1)	2.000	2.000
2nd Quartile (Q2, Median)	4.000	4.000
3rd Quartile (Q3)	6.000	6.000
Interquartile Range (IQR)	4.000	4.000

Table 3: Descriptive statistics of numeric variables (`score1` and `score2`).

3.2 Categorical Variables

Category	**Count**	**Proportion (%)**
Win (result = W)	54365	53.7549
Loss (result = L)	46735	46.2105
Tie (result = T)	35	0.0346

Table 4: Counts and proportions for the variable `result`.

Team1	**Count**	**Proportion (%)**
MIN	3548	3.5082
CHN	3546	3.5062
PHI	3544	3.5042
SFN	3542	3.5022
TEX	3542	3.5022
BOS	3540	3.5003
SDN	3538	3.4983
CIN	3538	3.4983
OAK	3536	3.4963
PIT	3535	3.4953
LAN	3535	3.4953
SLN	3534	3.4943
HOU	3533	3.4934
DET	3533	3.4934
BAL	3530	3.4904
NYN	3528	3.4884
SEA	3527	3.4874
MIL	3527	3.4874
TOR	3527	3.4874
NYA	3526	3.4864
KCA	3524	3.4845
CHA	3524	3.4845
CLE	3522	3.4825
ATL	3521	3.4815
COL	2509	2.4808
ANA	2221	2.1961
ARI	2136	2.1120
TBA	2136	2.1120
MON	1946	1.9242
WAS	1570	1.5524
FLO	1501	1.4842
CAL	1322	1.3072
MIA	994	0.9828

Team2	**Count**	**Proportion (%)**
TOR	3541	3.5013
CHA	3541	3.5013
SEA	3540	3.5003
MIL	3540	3.5003
HOU	3538	3.4983
ATL	3537	3.4973
SDN	3537	3.4973
NYA	3536	3.4963
KCA	3535	3.4953
LAN	3535	3.4953
CLE	3535	3.4953
CIN	3534	3.4943
NYN	3532	3.4924
BAL	3532	3.4924
DET	3531	3.4914
OAK	3531	3.4914
SFN	3528	3.4884
SLN	3527	3.4874
BOS	3526	3.4864
PHI	3524	3.4845
TEX	3524	3.4845
PIT	3523	3.4835
MIN	3520	3.4805
CHN	3515	3.4756
COL	2511	2.4828
ANA	2213	2.1882
ARI	2136	2.1120
TBA	2134	2.1101
MON	1983	1.9607
WAS	1566	1.5484
FLO	1509	1.4921
CAL	1313	1.2983
MIA	1008	0.9967

Table 5: Counts and proportions for team identifiers (`team1` and `team2`)

Team	**Count**	**Proportion (%)**
SDN	7075	3.4978
CIN	7072	3.4963
HOU	7071	3.4958
LAN	7070	3.4953
SFN	7070	3.4953
TOR	7068	3.4943
MIN	7068	3.4943
PHI	7068	3.4943
SEA	7067	3.4938
MIL	7067	3.4938
OAK	7067	3.4938
BOS	7066	3.4934
TEX	7066	3.4934
CHA	7065	3.4929
DET	7064	3.4924
BAL	7062	3.4914
NYA	7062	3.4914
SLN	7061	3.4909
CHN	7061	3.4909
NYN	7060	3.4904
KCA	7059	3.4899
ATL	7058	3.4894
PIT	7058	3.4894
CLE	7057	3.4889
COL	5020	2.4818
ANA	4434	2.1921
ARI	4272	2.1120
TBA	4270	2.1110
MON	3929	1.9425
WAS	3136	1.5504
FLO	3010	1.4881
CAL	2635	1.3027
MIA	2002	0.9898

Table 6: Counts and proportions for team identifiers (occurrences in `team1` and `team2` combined)

4 Visualizations

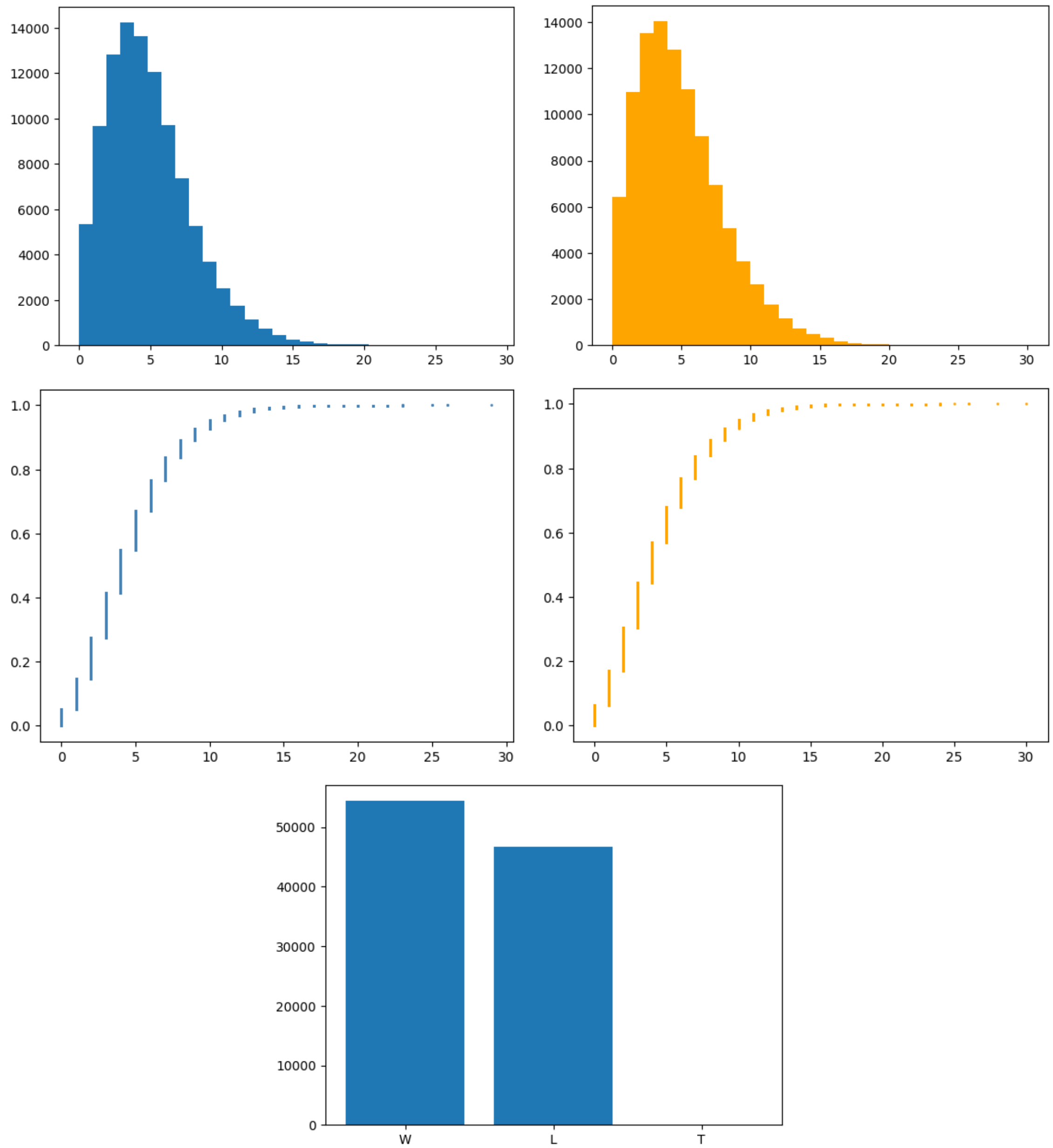

Figure 1: Descriptive visualizations of MLB game statistics: (a) Histogram of Score1, (b) Histogram of Score2, (c) ECDF of Score1, (d) ECDF of Score2, (e) Distribution of game results.

5 Discussion

The MLB dataset has 101,135 rows and 8 columns, which record every game in the reagular season from 1980 to 2024, and there are no missing values. Table 2 shows the variable in each column represents and its data type.

Table 3 and 4 are the descriptive statistics of `score1`, `score2` (numerical) and `result` (categorical). **For the scores, home teams have a higher mean and less variance, this corresponds to there are more wins in the `result` column (which is on the home team's view).** Table 5 and 6 counts the frequency that each team plays.

The Figures shows the distribution of the scores. Note that they generally have the same distribution with a right skewed tail, and they reach their mode at 3. However, due to the large sample size, I believe it is still reasonable to make some high-power statistical testing on whether home team always have a higher chance to win the game.

6 References

Data source: MLB dataset (mlb_data.csv).
Available at: https://www.retrosheet.org/gamelogs/index.html.